



**AI COMPUTER VISION
DKA3223**

TAJUK :

AI CODE ASSISTANT MINI PROJECT

KELAS / PROGRAM :
2 DKA (2026)

NAMA AHLI KUMPULAN :

AMEER DANISH FIRDAUS BIN AMEER FAEZAL	
MUHAMMAD AMIRUL 'AMMAR BIN NORIZAM	

NAMA PENSYARAH :
TS SARASAWATHY A/P CHANDIRAN

AI VISION PROJECT

COMPARING AI CODE ASSISTANTS

COMPARE • EVALUATE • RECOMMEND

COURSE
AI Vision

TOPIC
AI Code Assistants

DURATION
45 – 60 minutes

WORK MODE
Pairs

OBJECTIVES

At the end of this project, you will be able to:

- 1 Use AI code assistants to generate programming code.
- 2 Compare the quality of code produced by different AI tools.
- 3 Evaluate the strengths and limitations of AI-generated code.
- 4 Recommend the most suitable AI code assistant based on evidence.

INSTRUCTIONS

- 1 Work in pairs.
- 2 Choose TWO AI code assistants from the list below (or any other AI coding tools your teacher approves).

ChatGPT

Google Gemini

Microsoft Copilot

Cursor AI

GitHub Copilot

- 3 Use the SAME prompt (shown below) in both tools.
- 4 Compare the outputs based on the criteria provided.
- 5 Complete all sections and submit your report.

THE PROMPT

Create a Python program that calculates the Body Mass Index (BMI). The program should ask the user to enter weight (kg) and height (m), calculate the BMI, display the BMI value rounded to two decimal places, and classify the result as Underweight, Normal, Overweight, or Obese.

COMPARE THE RESULTS

CRITERIA	AI TOOL 1 ()	AI TOOL 2 ()	COMMENTS (Your Observations)
✓ Generated correct code	☐ Yes ☐ No	☐ Yes ☐ No	
▶ Code runs without errors	☐ Yes ☐ No	☐ Yes ☐ No	
💡 Code is easy to understand	☆ ☆ ☆ ☆ ☆	☆ ☆ ☆ ☆ ☆	
💬 Code includes comments	☆ ☆ ☆ ☆ ☆	☆ ☆ ☆ ☆ ☆	
</> Code is well organised	☆ ☆ ☆ ☆ ☆	☆ ☆ ☆ ☆ ☆	
📖 AI explains the code clearly	☆ ☆ ☆ ☆ ☆	☆ ☆ ☆ ☆ ☆	
⌚ Response speed	☆ ☆ ☆ ☆ ☆	☆ ☆ ☆ ☆ ☆	

RATING GUIDE:

☆☆☆☆ = Poor
☆☆☆☆ = Average
☆☆☆☆ = Excellent

REFLECTION QUESTIONS

- 1 Which AI tool generated better code?
- 2 Why do you think it was better?
- 3 Did both tools produce the same answer? Why or why not?
- 4 Which tool would you use for programming assignments? Explain your answer.
- 5 What is one limitation of using AI code assistants?

CONCLUSION

Write a short conclusion (100–150 words) summarising your comparison and stating which AI code assistant you recommend and why.

ASSESSMENT RUBRIC (20 MARKS)

CRITERIA	MARKS
Correct use of both AI tools	4
Comparison table completed	6
Quality of analysis	6
Reflection and conclusion	4
TOTAL	20

BONUS CHALLENGE (OPTIONAL)

Improve the prompt to get a better result. For example:

"Create a Python BMI calculator with input validation, comments, error handling, and coloured output."

Then compare the new results.

AI CHATBOT DIPILIH: CHATGPT & GEMINI AI

CHATGPT AI:

PROMPT (SAMA)

Create a Python program that calculates the Body Mass Index (BMI). The program should ask the user to enter weight (kg) and height (m), calculate the BMI, display the BMI value rounded to two decimal places, and classify the result as Underweight, Normal, Overweight, or Obese.

AI TOOL 1 (ChatGPT)

Python



Run

```
weight = float(input("Enter your weight (kg): "))
height = float(input("Enter your height (m): "))

bmi = weight / (height ** 2)

print(f"\nYour BMI is {bmi:.2f}")

if bmi < 18.5:
    print("Category: Underweight")
elif bmi < 25:
    print("Category: Normal")
elif bmi < 30:
    print("Category: Overweight")
else:
    print("Category: Obese")
```



ChatGPT_Code.ipynb



File Edit View Insert Runtime Tools Help

Q Commands + Code + Text ▶ Run all



[]



```
weight = float(input("Enter your weight (kg): "))
height = float(input("Enter your height (m): "))

bmi = weight / (height ** 2)

print(f"\nYour BMI is {bmi:.2f}")

if bmi < 18.5:
    print("Category: Underweight")
elif bmi < 25:
    print("Category: Normal")
elif bmi < 30:
    print("Category: Overweight")
else:
    print("Category: Obese")
```

give data to test the categories.

Here are some test data you can use to verify all four BMI categories.

Category	Weight (kg)	Height (m)	BMI	Expected Output	
Underweight	45	1.70	15.57	Underweight	
Normal	60	1.70	20.76	Normal	
Overweight	75	1.70	25.95	Overweight	
Obese	95	1.70	32.87	Obese	

ChatGPT_Code.ipynb ☆

File Edit View Insert Runtime Tools Help

Commands + Code + Text ▶ Run all

[3] ✓ 7s

▶

```
weight = float(input("Enter your weight (kg): "))
height = float(input("Enter your height (m): "))

bmi = weight / (height ** 2)

print(f"\nYour BMI is {bmi:.2f}")

if bmi < 18.5:
    print("Category: Underweight")
elif bmi < 25:
    print("Category: Normal")
elif bmi < 30:
    print("Category: Overweight")
else:
    print("Category: Obese")
```

⌵

... Enter your weight (kg): 45
Enter your height (m): 1.70

Your BMI is 15.57
Category: Underweight

ChatGPT_Code.ipynb ☆

File Edit View Insert Runtime Tools Help

Commands + Code + Text ▶ Run all

[4] ✓ 7s

▶

```
weight = float(input("Enter your weight (kg): "))
height = float(input("Enter your height (m): "))

bmi = weight / (height ** 2)

print(f"\nYour BMI is {bmi:.2f}")

if bmi < 18.5:
    print("Category: Underweight")
elif bmi < 25:
    print("Category: Normal")
elif bmi < 30:
    print("Category: Overweight")
else:
    print("Category: Obese")
```

⌵

... Enter your weight (kg): 60
Enter your height (m): 1.70

Your BMI is 20.76
Category: Normal

[5]

✓ 9s

```
weight = float(input("Enter your weight (kg): "))
height = float(input("Enter your height (m): "))

bmi = weight / (height ** 2)

print(f"\nYour BMI is {bmi:.2f}")

if bmi < 18.5:
    print("Category: Underweight")
elif bmi < 25:
    print("Category: Normal")
elif bmi < 30:
    print("Category: Overweight")
else:
    print("Category: Obese")
```

▼

```
... Enter your weight (kg): 75
Enter your height (m): 1.70
```

```
Your BMI is 25.95
Category: Overweight
```

[6]

✓ 5s

```
weight = float(input("Enter your weight (kg): "))
height = float(input("Enter your height (m): "))

bmi = weight / (height ** 2)

print(f"\nYour BMI is {bmi:.2f}")

if bmi < 18.5:
    print("Category: Underweight")
elif bmi < 25:
    print("Category: Normal")
elif bmi < 30:
    print("Category: Overweight")
else:
    print("Category: Obese")
```

▼

```
... Enter your weight (kg): 95
Enter your height (m): 1.70
```

```
Your BMI is 32.87
Category: Obese
```

GEMINI AI:

Prompt dan jawapan AI:

ates the Body
enter weight (kg)
to decimal plac
or Obese.

buat ni di dalam bahasa phyton.
saya tulis kod ni guna google collab ye

Boleh, ini kod Python yang lengkap dan sedia untuk dijalankan di Google Colab.

Anda hanya perlu salin (copy) kod di bawah dan tampal (paste) ke dalam code cell di Google Colab.

Kod Python Calculator BMI

```
Python
# 1. Menerima input berat (kg) dan tinggi (m) daripada pengguna
berat = float(input("Masukkan berat badan anda (kg): "))
tinggi = float(input("Masukkan tinggi anda dalam meter (m): "))

# 2. Menghitung BMI (Formula: Berat / Tinggi^2)
bmi = berat / (tinggi ** 2)

# 3. Menentukan kategori berdasarkan nilai BMI
if bmi < 18.5:
    kategori = "Underweight"
elif 18.5 <= bmi < 25:
    kategori = "Normal"
elif 25 <= bmi < 30:
    kategori = "Overweight"
else:
    kategori = "Obese"

# 4. Memaparkan keputusan (BMI dibundarkan kepada 2 tempat perpuluhan)
print("-" * 35)
print(f"Nilai BMI anda : {bmi:.2f}")
print(f"Kategori      : {kategori}")
print("-" * 35)
```

Hasil kod:

Data yang digunakan:

boleh beri data input untuk dapat semua kategori?

Boleh, sangat boleh!

Berikut adalah data ujian (*test data*) yang boleh anda masukkan satu per satu ke dalam program Python BMI tersebut untuk menguji dan mendapatkan **keempat-empat kategori** hasil:

Senarai Input Mengikut Kategori

Kategori BMI	Contoh Berat (kg)	Contoh Tinggi (m)	Pengiraan BMI
Underweight	45	1.70	15.57
Normal	65	1.70	22.49
Overweight	80	1.70	27.68
Obese	95	1.70	32.87

Underweight

```
Masukkan berat badan anda (kg): 45
Masukkan tinggi anda dalam meter (m): 1.70
-----
Nilai BMI anda : 15.57
Kategori      : Underweight
-----
```

Normal

```
Masukkan berat badan anda (kg): 65
Masukkan tinggi anda dalam meter (m): 1.70
-----
Nilai BMI anda : 22.49
Kategori      : Normal
-----
```

Overweight

```
Masukkan berat badan anda (kg): 80
Masukkan tinggi anda dalam meter (m): 1.70
-----
Nilai BMI anda : 27.68
Kategori      : Overweight
-----
```

Obese

```
Masukkan berat badan anda (kg): 95
Masukkan tinggi anda dalam meter (m): 1.70
-----
Nilai BMI anda : 32.87
Kategori      : Obese
-----
```

COMPARE THE RESULT

CRITERIA	AI TOOL 1 (Gemini)	AI TOOL 2 (ChatGPT)	COMMENTS (Your Observations)
Generated correct code	Yes	Yes	Both generated correct BMI calculator. (chatgpt comment)
Code runs without errors	Yes	Yes	Both programs executed successfully (chatgp comment).
Code is easy to understand	☆ ☆ ☆ ☆	☆ ☆ ☆ ☆ ☆	ChatGPT uses clearer formatting. (chatgpt comment)
Code includes comments	☆ ☆ ☆ ☆ ☆	☆ ☆ ☆ ☆ ☆	Neither included comments in the basic prompt. (chatgpt comment)
Code is well organised	☆ ☆ ☆ ☆ ☆	☆ ☆ ☆ ☆ ☆	ChatGPT has cleaner layout and spacing (chatgpt comment).
AI explains the code clearly	☆ ☆ ☆ ☆ ☆	☆ ☆ ☆ ☆ ☆	ChatGPT provides more detailed explanations. (chatgpt comment)
Response speed	☆ ☆ ☆	☆ ☆ ☆ ☆ ☆	ChatGPT responded slightly faster. (chatgpt comment)

REFLECTION QUESTIONS:

1. Which AI tool generated better code?

ChatGPT generated better code because it was more organised and easier to understand.

2. Why do you think it was better?

The code used clear formatting, meaningful variable names, and easy-to-follow logic. ChatGPT also explained how the code works in a simple way.

3. Did both tools produce the same answer? Why or why not?

Yes. Both tools calculated the BMI correctly and classified the user's BMI into the correct category. The only difference was the coding style and formatting.

4. Which tool would you use for programming assignments? Explain your answer.

I would use ChatGPT because it provides accurate code, clear explanations, and well-organised solutions. It also makes it easier to learn programming concepts.

5. What is one limitation of using AI code assistants?

AI code assistants may sometimes generate incorrect or inefficient code, so users should always test and understand the code before using it.

CONCLUSION

This project compared ChatGPT and Google Gemini using the same Python BMI calculator prompt. Both AI tools successfully generated working programs that calculated BMI and displayed the correct classification. However, ChatGPT produced code that was better organised, easier to read, and supported by clearer explanations. Gemini also generated correct code, but its response was slightly less detailed. Based on the comparison, I recommend ChatGPT for programming assignments because it produces accurate code, provides helpful explanations, and is easier for beginners to understand. Although AI code assistants are useful for learning and improving productivity, users should still review, test, and understand the generated code instead of relying on it completely.